

Royal Netherlands Institute for Sea Research

FAIR - UNECSS workshop

Best Practices for Making (Meta)Data FAIR

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What does interoperable mean?

Principles

- I1. (Meta)data use a formal, accessible, shared & broadly applicable language for knowledge representation
- I2. (Meta)data use vocabularies that follow FAIR principles
- I3. (Meta)data include qualified references to other (meta)data

What does interoperable mean?

Principle

- I1. (Meta)data use a formal, accessible, shared & broadly applicable language for knowledge representation



Data needs to be understood by humans & machines

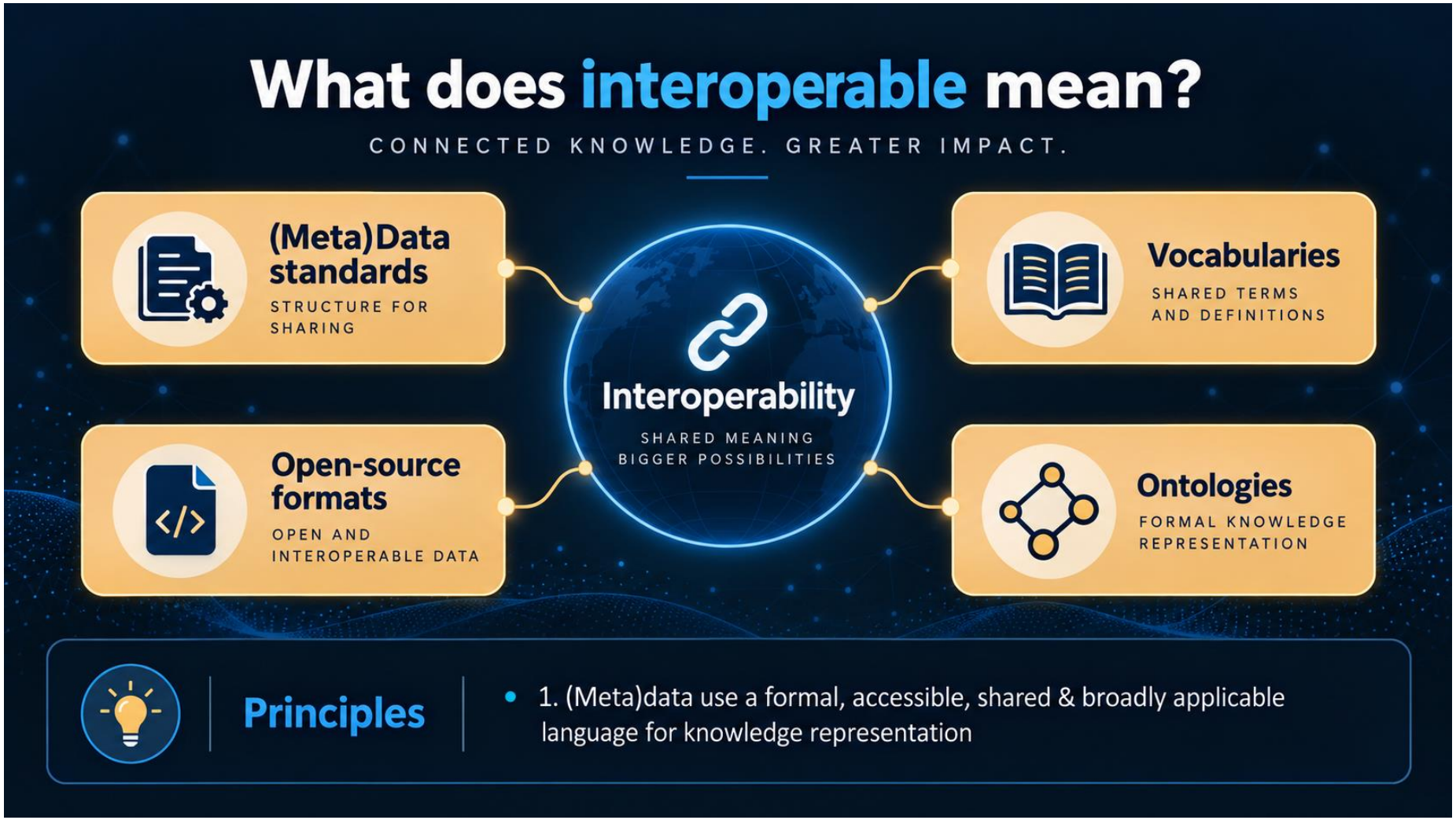
What does interoperable mean?



Principles

- I1. (Meta)data use a formal, accessible, shared & broadly applicable language for knowledge representation
  - There are different ways in which you can describe your data --> Avoid ambiguity!
  - Terminology might be highly domain-specific and therefore difficult to **understand**

What does interoperable mean?



What does interoperable mean?

Principle

- I2. (Meta)data use vocabularies that follow FAIR principles

Controlled vocabulary

= A controlled vocabulary is a carefully selected list of terms which have well-defined meanings, and which tag specific units of information: each term in the vocabulary means just one thing.

- It can help to restrict the terminology that you are using to describe your data to previously defined terms.
- It ensure standardisation

What does interoperable mean?

Principle

- I3. (Meta)data include qualified references to other (meta)data

Qualified reference

= cross-reference that explains its intent

- For example, 'X is regulator of Y' instead of 'X see also Y'
- Create meaningful links between (meta)data to enrich the contextual knowledge about the data
- Specify when one datasets builds on another data set, if additional datasets are needed to complete the data or if complementary information is stored in different dataset.
- datasets need to be properly cited (i.e., including their globally unique and persistent identifiers)

What makes (meta)data interoperable?

Interoperable (meta)data can be discovered, combined, and reused across systems and workflows, increasing their visibility, usability, and impact.

Interoperable (meta)data requires:

- Standard schema's
- Standard models (**standard data models / knowledge-representation models)**
- Standard formats
- Provided in open formats: if you cannot open, you cannot operate on
- Self-contained, self-explanatory

Schemas examples:

Ecological Metadata Language (EML), Dublin Core, DCAT (Data Catalog Vocabulary), Darwin Core (Dwc), DataCite, CERIF, etc.

Controlled vocabularies

… requires the use of:

- Controlled vocabularies (PIDs/URLs/DOIs)
  - people (ORCiD), institutes (ROR)
  - Protocols and SOPs, documents, related digital objects (DOI, URLs)
  - instrument, parameters, units (BODC NVS, PIDs)
  - taxonomy, locations (WoRMS, MarineRegions, NCBI)
  - descriptive terms (ENVO)
  - …

Every term is unambiguous: everyone can understand it

Controlled vocabularies

Both also require the use of

- Standard formats for strings and values
  - decimals, units (3.45 or 3,45? mum, micrometre, mircometER, or a ...?)
  - dates, locations (YYYY-MM-DD or YYYY-DD-MM? No or Norway?)
  - missing values (NA, NAN, NC, blank?)
  - a convention for column titles/metadata fields and values (water temperature,, temp, temperature (sea water), https://vocab.nerc.ac.uk/collection/P33/current/WATERTEMPERATURE/?) → semantics
  - ...

* Semantics = the meaning of a word, phrase, or text

Every term is unambiguous: everyone can understand it

Interoperable (meta)data content

- Data: Column titles, acronyms, etc are explained
  - Temp ss = http://vocab.nerc.ac.uk/collection/A04/current/SeaSurfaceTemperature/
- Data: Units are included as separate values rather than implied or written together
  - ~~| 20 C |~~ | 20 | Celsius |
- (Meta)data: Controlled vocabularies are used for terms
  - "T ss", "Temp ss" “sea surface temperature” ?
  - Unambiguous
    - URL http://vocab.nerc.ac.uk/collection/A04/current/SeaSurfaceTemperature/
    - PID: SDN:A04::SeaSurfaceTemperature
  - URL always in metadata; PID can be used in data BUT needs additional information in the README or metadata.csv file
- (Meta)data: Standardized and defined formats are used for data values
  - YYYYMMDD or YYYYDDMM? → Recommended YYYY-MM-DD
  - XKCD cartoon
- ISO country formats: NO or NOR

Interoperable spreadsheets

- *Spreadsheets* .xls **no** .csv/.tsv **yes**

Excel: Why using Microsoft's tool caused Covid-19 results to be lost

The badly thought-out use of Microsoft's Excel software was the reason nearly 16,000 coronavirus cases went unreported in England.

Source: https://www.bbc.com/news/technology-54423988

Interoperable spreadsheets

An interoperable spreadsheet is regular, standardised, and text-based

- Formatted as CSV: *nothing is lost in translation*
- Rectangular and standardised rows+columns: a*llows easy programmatic access*
  - Start at the top-left and fill in completely until the bottom-right
  - One table per spreadsheet
- Provide a separate metadata spreasheet with all your terms, abbrieviations, titles explained
- Provide a README.txt with background info

Interoperable spreadsheets

Data.csv

date	temp
2022-02-01	23
2022-02-02	23
2022-02-03	18
2022-02-04	12
2022-02-05	10

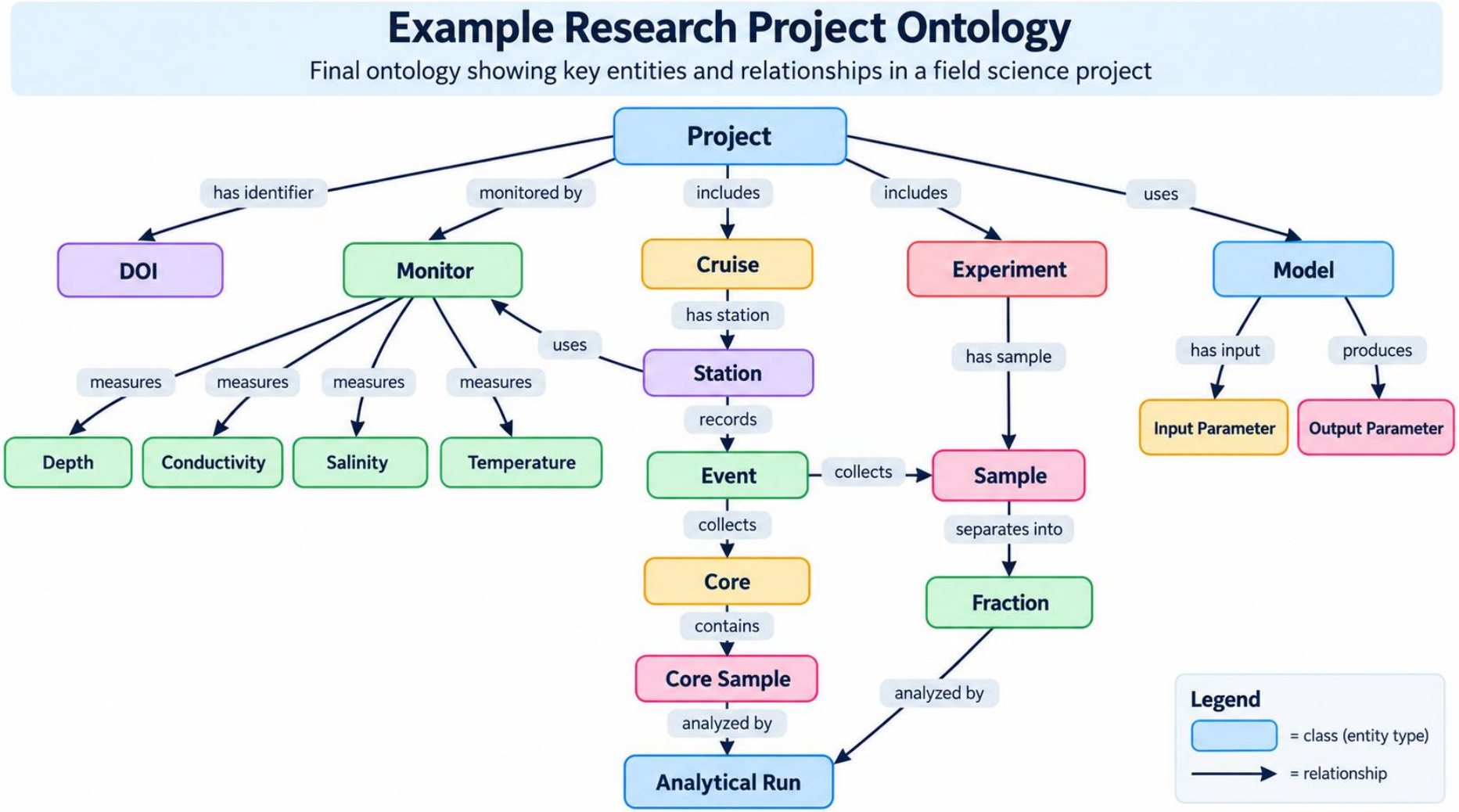
Metadata.csv

term	description	unit	term property	unit property
date	date the ob	none	schema:Date	none
temp	water temp	celsius	http://vocab.nerc.ac.uk/collection/P02/current/TEMP/	http://vocab.nerc.ac.uk/collection/P06/current/UPAA/

Machine-readable metadata formats

- Standard formats
  - XML (eXtensible Markup Language)
  - TTL (Terse RDF Triple Language)
  - JSON (JavaScript Object Notation) and JSON-LD (JavaScript Object Notation for Linked Data)
- Metadata standards
  - https://en.wikipedia.org/wiki/Metadata_standard for a list
  - Biology/ecology: Darwin Core, EML...see this DCC list
  - Genomics: MIxS
  - Oceanography: see this DCC list
  - Provenance: PROV-DM
  - Dataset descriptions: DCAT

Ontology


Example Research Project Ontology
Final ontology showing key entities and relationships in a field science project
Project
has identifier
monitored by
includes
includes
uses
DOI
Monitor
Cruise
Experiment
Model
has station
has sample
has input
produces
uses
measures
measures
measures
measures
Station
Input Parameter
Output Parameter
Depth
Conductivity
Salinity
Temperature
records
Event
collects
Sample
collects
separates into
Core
Fraction
contains
Core Sample
analyzed by
analyzed by
Analytical Run
Legend
= class (entity type)
= relationship

Machine-readable metadata formats

How is this relevant to me?

- Every dataset (possibly every data *file*) should have a README/metadata file to describe the data
- At a minimum this should be written either as a spreadsheet (interoperable) or a plain text file
- Ideally should be written in an interoperable format -> does require specialised knowledge

QUESTIONS AND HANDS ON

